Memora

Decision Intelligence Platform

Local-First Context Engine for High-Stakes Decisions

Executive Project Proposal

System Architecture & Strategic Vision

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1 Executive Summary

Key Insight

Memora is a local-first decision intelligence platform that turns everything a person captures—text, voice, images—into a structured, interconnected knowledge graph. Three specialized AI agents continuously reason over this living graph to surface hidden connections, flag neglected commitments, recommend optimal actions, and deliver real-time situational awareness that no human memory can maintain.
This is not a note-taking app. Context capture is the input; **better high-stakes decisions, proactive recommendations, and strategic foresight** are the output.

1.1 The Problem

- **67% of saved notes are never revisited**—current knowledge tools are passive graveyards.
- **No tool models life holistically**—calendar, CRM, task manager, journal—none talk to each other.
- **Cross-domain connections are invisible**—stress correlates with overcommitment correlates with declining grades, but no tool sees this.
- **AI design determines outcomes**—Harvard/MIT research shows the same AI can improve performance by +1.3 standard deviations or *harm* it by –0.5 standard deviations, depending on implementation.

1.2 The Opportunity

Persistent memory is the market differentiator (Mem0: $24M raise, 41K GitHub stars). But no production system combines: local-first privacy, self-building knowledge graphs, cross-domain reasoning, fact-checked intelligence, internet enrichment, temporal knowledge evolution, and **proactive decision recommendations**. Memora is the first to attempt all seven.

1.3 Proposed Timeline & Cost

Phase	Duration	Scope
Foundation + Archivist + Truth Layer	4–5 weeks	Core graph, extraction pipeline, fact verification
Strategist + Researcher + Daily Briefing	3–4 weeks	Cross-network analysis, internet research, briefings
Web Frontend	5–7 weeks	Capture, review, graph viz, network dashboard
Total MVP	**12–16 weeks**	

Infrastructure cost: **$0** (embedded databases). User-facing LLM cost: **$10–15/month** BYOK.

2 Context & Background

2.1 Why "Decision Intelligence" Not "Life OS" or "Second Brain"

The "Life OS" framing invites comparison to Notion templates—productivity theater. "Second Brain" implies passive storage. Memora is neither. The actual value proposition:

> *"I need to make a major decision today. Give me the full picture—from my personal context AND from the world—then tell me what I'm missing and what I should do." No tool does this.*

Memora goes beyond context capture into **active decision support**: recommending actions, identifying risks before they materialize, surfacing strategic opportunities across domains, and challenging your assumptions with adversarial thinking on demand.

Real-world examples:

- **Running an election**: relationship graphs + opposition research + endorsement chains + *"Who should you call today and why?"*
- **Negotiating a deal**: history with counterparty + market comparables + past commitments + *"Based on your leverage, here's your optimal opening position"*
- **Research strategy**: hypothesis evolution + latest papers + collaborator connections + *"This new paper contradicts your assumption from March—reconsider?"*
- **Startup operations**: product decisions + competitor launches + user feedback + *"You're overcommitted: 4 deadlines in 3 days, recommend deferring X"*

2.2 Market Context

The PKM (Personal Knowledge Management) space is mature but fragmented. $4.5B+ has been invested across players like Notion (100M+ users), Tana ($25M at $100M post-money valuation), Obsidian (~$2M ARR bootstrapped), and Heptabase (350K users, $7M ARR). Yet **every tool stops at capture**. None reason across domains. None verify facts. None model temporal evolution.

2.3 Lessons from the Graveyard

Product	Lesson Learned
Dot (New Computer)	Shut down Sept 2025. AI companion without utility $\neq$ PMF. Memora must be a *tool*, not a *friend*.
Limitless / Rewind	Acquired by Meta Dec 2025. Users lost data permanently. Local-first must mean user *owns* the data files.
Roam Research	Pioneered bidirectional linking, slow decline since 2021. Category creators must keep iterating.
Mem.ai	Called "$40M failure." No proprietary data structures. Pure wrapper risk realized.
Jasper ($125M raised)	Lost PMF when primary workspaces added native AI. Adjacent tools die when the primary workspace adds AI.

3 Vision & Core Principles

3.1 The Input Model

Two input types only: text and images. No calendar sync, no bank feeds, no location tracking. Everything comes from the user describing their life in their own words.

- Voice memos transcribed via Whisper to text
- Screenshots processed via OCR + LLM to text
- **You are the sole source of truth**—the system never pulls data silently

3.2 The Graph Construction Model

LLM-driven, human-validated. The LLM proposes graph changes. Nothing enters the graph without approval.

Core data flow: every piece of information follows the capture–propose–review–commit pipeline. The Archivist extracts structured entities and relationships from natural language input, proposes typed graph changes via constrained Pydantic schemas, and presents them for human review.

3.3 Validation Modes

Decision fatigue is real—quality degrades after ~35 micro-decisions per day (Baumeister et al.). The system defaults to **auto-approve at ≥85% confidence** from day one, with a daily review digest for corrections.

Mode	Trigger	UX
Auto-approve	$\geq$ 0.85 confidence	Silently committed; shown in daily digest
Daily review digest	All auto-committed	Review what was committed, correct mistakes
Explicit confirm	High-impact changes	Deleting nodes, resolving contradictions

Success Metric

Correction rate target: <5% of auto-approved changes require user correction, indicating a well-calibrated confidence threshold.

4 System Architecture Overview

4.1 High-Level Context Diagram

The following diagram shows Memora as a system boundary, with external actors and data flows at the business level. This is the "30,000-foot view" — one system, its users, and what crosses the boundary.

System context diagram. Memora is a single system boundary that accepts user captures, retrieves verified information from the internet, leverages LLM providers for extraction and reasoning, and delivers decisions, recommendations, strategic insights, and daily briefings back to the user across devices.

4.2 Layered Architecture

Four-layer architecture. All AI intelligence is grounded in deterministic algorithms (decay, health scoring, bridge discovery) and verified facts (Truth Layer), ensuring the LLM augments but never replaces the underlying data structures.

5 Context Networks

5.1 The Core Concept

The graph is organized into **multiple interconnected context networks**—living subgraphs that each track a domain of the user's life with their own health status, momentum, and alerts. A single node can belong to multiple networks, and the **cross-network bridges** between them are where the highest-value insights emerge.

Seven default context networks organized around the user's life graph. Cross-network bridges (gold arrows) represent the platform's primary value: surfacing connections invisible to any single-domain tool. Example: a grade decline (Academic) correlating with a new client onboarding (Professional) triggers a proactive alert.

5.2 Network Health

Each network maintains a health status derived from graph state—never a misleading numeric score.

Status	What It Means
On Track	Commitments met, no overdue items, recent engagement
Needs Attention	Some open alerts, declining momentum, stale commitments
Falling Behind	Multiple overdue items, no recent input, high alert count

Health is computed from **commitment completion rate** (actionable), **alert ratio** (urgency), and **staleness flags** (only triggered when commitments exist but haven't been updated—silence is fine when there are no deadlines). Momentum is internal-only as a directional arrow (up/stable/down)—never shown as a percentage, because 7-day self-reported trends are too noisy to quantify.

6 The AI Council

6.1 Why Three Agents, Not Five

Research from Google DeepMind (Dec 2025) shows that independent multi-agent systems amplify errors by **17.2**×. Accuracy saturates at 4–5 agents. For sequential reasoning, single agents with better tools are strictly superior. Graph-mesh topology yields the best results; unstructured "bag of agents" is the worst.

The council was reduced from 5 agents to 3, each with a clear, non-overlapping role:

6.2 Council Decision Pattern

For high-stakes queries, the council follows a structured deliberation process:

1. **Decomposition**: Orchestrator breaks the problem into domain-specific sub-questions
2. **Independent Analysis**: Each relevant agent analyzes in parallel
3. **Proposal Submission**: Each submits answer, confidence (0–1), evidence, and risks
4. **Deliberation** (optional, high-stakes only): Agents review each other's proposals (max 2–3 rounds)
5. **Synthesis**: Confidence-weighted aggregation. High disagreement flagged for human review

6.3 LLM Routing & Cost

Agent	Recommended Model	Rationale
The Archivist	Claude Haiku	High-frequency, structured output, cost-sensitive
The Strategist	Claude Sonnet	Low-frequency, complex reasoning
The Researcher	Claude Sonnet	Low-frequency, synthesis from web sources

With prompt caching (0.1× base cost for cache hits), the Archivist's costs drop 60–70%. Blended user cost: **$10–15/month** at 10 captures/day and 2 council queries.

7 Data Architecture

7.1 Zero-Infrastructure Storage

Both databases are **embedded**—they run inside the application process with no external servers, containers, or cloud services. Total infrastructure cost: **$0**.

7.2 No Separate Memory System

The graph **is** the memory. The knowledge graph with embeddings already provides persistent memory, entity resolution, and temporal tracking—exactly what standalone memory systems (Mem0, Zep) do. Adding a separate memory layer would create two sources of truth. The graph subsumes all memory functions.

8 The Truth Layer: Verified Fact Store

8.1 The Problem It Solves

The graph contains user-reported information, LLM-extracted inferences, and internet research findings. Without verification, the graph accumulates unverified claims that agents reason over as fact. **If the Strategist recommends an action based on a hallucinated inference from a stale article, you've made a decision on fiction.**

8.2 Architecture

8.3 Source Confidence Hierarchy

Source Type	Confidence	Examples
PRIMARY	Highest	Official records, government databases, published financials
SECONDARY	Medium	News articles, research papers, third-party reports (with citation)
SELF_REPORTED	Lowest	User-stated hearsay ("Sam told me X")—flagged as such

Facts with `DYNAMIC` expiry get periodic rechecking via the Researcher agent. Static facts (birth dates, published papers) never expire. Stale facts are flagged: *"This fact was last verified 30 days ago—recheck?"*

9 RAG Pipeline & Retrieval

Memora's retrieval system goes beyond standard vector search by combining graph traversal with semantic retrieval, routed adaptively based on query complexity.

Key Insight

CRAG (Corrective RAG): If the retrieval quality is poor, the system automatically falls back to web search via the Researcher agent rather than generating from insufficient context. This is how the private graph and public internet blend seamlessly.

10 Graph Schema Overview

The graph uses two categories of typed nodes—**Life Context Nodes** (events, people, commitments, decisions) and **Knowledge Nodes** (notes, ideas, concepts, references)—connected by seven categories of typed edges.

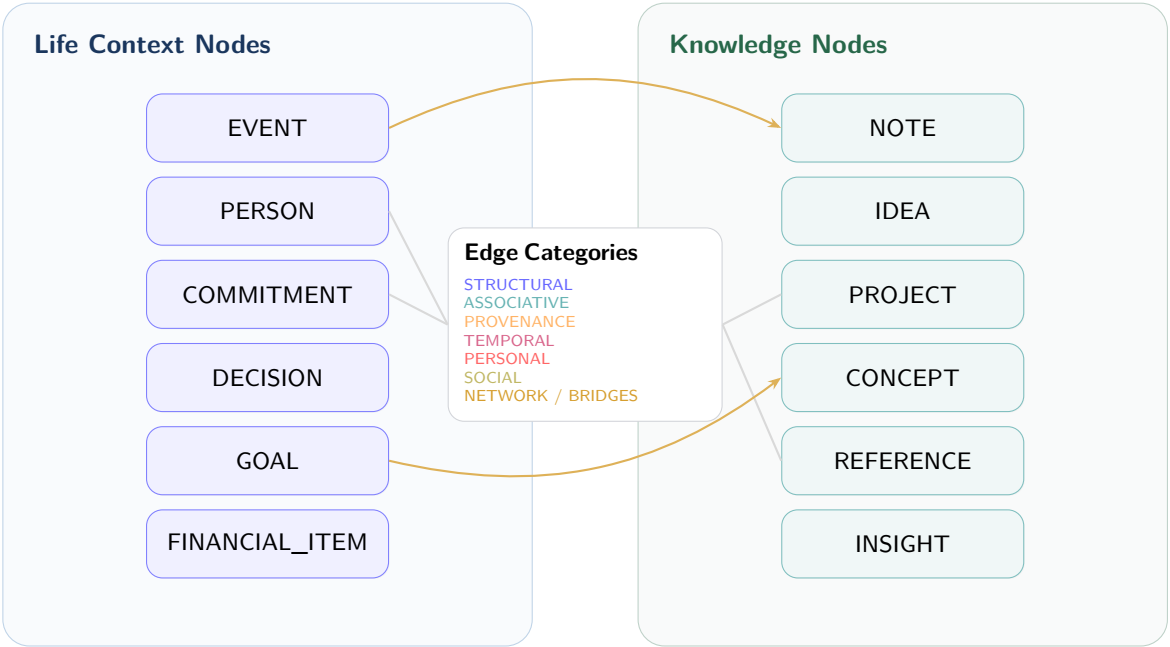

Figure 5: High-level graph schema. Every node carries embeddings, timestamps, confidence scores, network memberships, and human-approval status. Edges are typed across seven categories, enabling precise graph traversal for different query types. Cross-category connections (gold) represent the richest semantic links.

10.1 Shared Node Properties

Every node in the graph carries:

- Unique ID + content hash
- Created / updated timestamps
- Embedding vector (for semantic search)
- Confidence score (0–1)
- Network memberships list
- Human-approved flag
- Proposed-by attribution
- Access count (for decay)
- Tags and review date

11 Notification & Briefing System

All notifications are derived entirely from graph state. No external tracking, no invasive monitoring. The user closes the loop by telling the system what happened—"submitted the proposal" marks the commitment as resolved and stops notifications.

11.1 Notification Triggers

Trigger	Example
Deadline approaching	"Midterm in 2 days—you mentioned you haven't started Ch. 7–9"
Decaying relationships	"You haven't mentioned Sam in 14 days—you owed him an intro"
Stale commitments	"Invoice to Koo was due 3 days ago—still marked open"
Network health drop	"Your Academic network has had no input in 8 days"
Cross-network alerts	"You mentioned stress twice this week and have 4 open commitments"
Goal drift	"Memora MVP target is March 31—progress at 15%"
Review queue	"3 concepts due for spaced repetition review"

11.2 The Daily Briefing

Every morning, the Strategist compiles a life situation report from graph state:

- Per-network health status and momentum direction
- Open alerts ranked by urgency
- Cross-network correlations discovered in the last 24 hours
- Recommended priorities and **decision prompts** for the day
- Proactive suggestions: people to contact, deadlines to renegotiate, opportunities to pursue
- Spaced repetition review items

12 Background Mechanics

These are the deterministic, LLM-independent algorithms that run on schedule—the "engine room" that makes the graph a living system rather than a static document store.

Job	Frequency	Purpose
Network Health	Every 6 hours	Recalculate health categories for all context networks
Decay Scoring	Daily	Exponential decay on all nodes based on last access / mention
Bridge Discovery	Daily	Embedding similarity scan for cross-network connections
Commitment Scan	Daily	Check open commitments against due dates, schedule alerts
Relationship Decay	Weekly	Flag people with no interaction in configurable window
Daily Briefing	Daily	Strategist compiles life situation report
Spaced Repetition	Daily	Surface knowledge nodes due for review (SM-2 algorithm)
Gap Detection	Weekly	Find isolated concepts, stalled goals, orphaned nodes

Key Insight

Bridge Discovery performance: Per-capture incremental discovery uses one embedding lookup against an HNSW index = $O(\log N)$. With 10K nodes, this is ~1–5ms. Daily batch scans only process nodes modified in the last 24 hours and batch LLM evaluations into a single call ("Here are 15 potential connections—which are meaningful?")—1 API call instead of 15.

13 Anti-Wrapper Positioning

13.1 The Wrapper Test

Google VP Darren Mowry (Feb 21, 2026): *"Wrapping thin IP around Gemini or GPT-5 is a sign that a startup is not distinguishing itself."*

The test: if your LLM provider shut off your API key, would your product die?

Memora passes. The knowledge graph, Truth Layer, decay mechanics, health scores, bridge discovery, spaced repetition, and cross-network reasoning are **all non-LLM systems**. The LLM makes them accessible, but the intelligence lives in data structures and algorithms.

13.2 Differentiation Stack

Figure 6: Differentiation stack ordered by distance from commodity LLM wrappers. The top three layers (graph, Truth Layer, bridge discovery) represent maximum differentiation—capabilities that cannot be replicated by prompting alone. The bottom layer (council chat) is the wrapper-risk zone, mitigated by always showing graph provenance and verified fact citations.

13.3 Anti-Wrapper Design Rules

1. Every AI output must cite **graph nodes AND verified facts**. ChatGPT cannot do this.
2. The graph is the product, the LLM is the interface. Remove the LLM and the graph + Truth Layer + mechanics remain a queryable life database.
3. Custom algorithms > LLM prompts. Bridge discovery = embedding similarity + graph traversal. Health = weighted metrics. Decay = exponential functions. Spaced repetition = SM-2. All deterministic and reproducible.
4. Progressive disclosure of AI reasoning. Answer first. Collapsible: agents, confidence, graph paths, verified facts, disagreements.

5. Never let the LLM be the sole source of intelligence. Every insight must be verifiable by graph query + Truth Layer lookup.

14 Competitive Landscape

14.1 Market Map

Product	Local	Graph	Deep AI	Key Gap
Tana ($25M)	No	Yes	Multi-model	Manual schema, cloud-dependent, no life-level reasoning
Obsidian	Yes	No	Plugins	No AI extraction, no cross-domain
Notion (100M+)	No	No	3.0 agents	Workspace-level, not life-level
Anytype	Yes	Partial	No	Strongest local-first. No AI whatsoever
Heptabase	Partial	Spatial	Light	Visual-only, no structured graph
Mem.ai	No	No	Core	No proprietary data structures
Memora	**Yes**	**Yes**	**Yes**	**Combines all three**

14.2 Market Gaps Memora Fills

Gap	Memora's Position
Privacy-preserving intelligence	Local-first + BYOK LLM API + local embeddings
Self-building structured knowledge	LLM-constructed, human-validated graph (vs. Tana's manual Supertags)
Cross-domain decision context	Context networks with cross-network bridges
Temporal knowledge	`EVOLVED_INTO` edges, temporal diffs, conviction scores
Fact-checked intelligence	Truth Layer with source typing and staleness detection
Internet-enriched personal graph	Researcher agent + MCP + Truth Layer
Proactive decision support	Strategist agent recommends actions, challenges assumptions, surfaces opportunities

Key Insight

No production system combines all seven. The concept is genuinely novel if executed. Closest competitor Tana requires manual schema definition, is cloud-dependent, has no cross-domain life reasoning, no internet research, and no fact verification.

15 Technology Stack

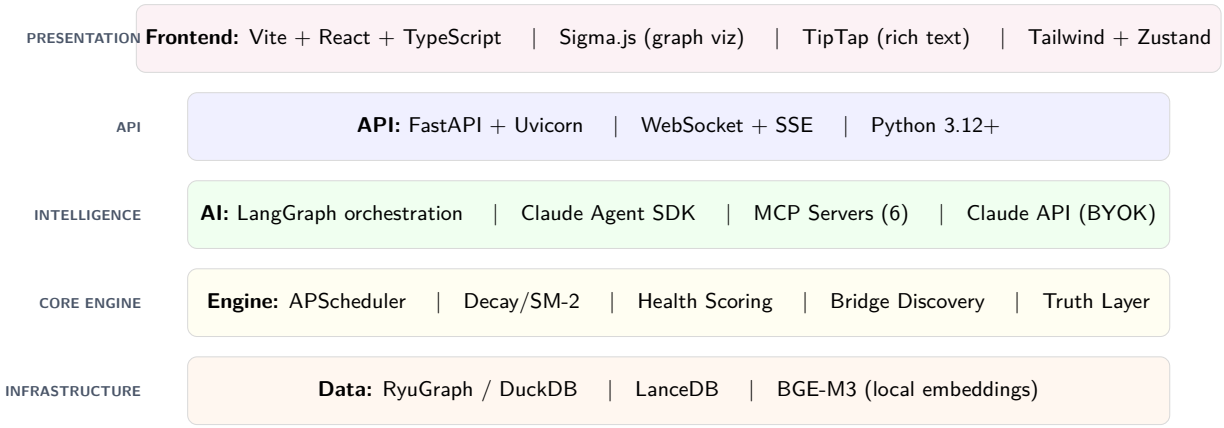

Figure 7: Technology stack overview. Each layer is independently testable and replaceable. The critical insight: the Core Engine layer is entirely LLM-independent—deterministic algorithms that would function even if the AI layer were removed.

15.1 MCP Servers (Agent Tools)

MCP Server	Purpose
Google Search	Primary web search (100 free/day)
Brave Search	Fallback (2,000 free/month)
Playwright	Full web scraping for deep research
Semantic Scholar + arXiv	Academic paper search and retrieval
GitHub MCP	Code and repository search
Graph + Vector DB (custom)	Query Memora as agent tools

16 UX Design Principles

1. **Graph as infrastructure, not interface**—the graph powers everything behind the scenes. Multiple views: graph, outline, timeline, table, network dashboard.
2. **Local graph > global graph**—default to neighborhood of current node. Global graphs become hairballs past a few hundred nodes.
3. **Command palette as primary navigation**—Cmd+K is fastest. Keyboard-first, mouse-friendly.
4. **Capture-first, structure-later**—every entry goes to the Archivist. You just talk naturally.
5. **Progressive disclosure of AI**—show the answer first. Collapsible sections reveal reasoning, sources, agent deliberation.
6. **Decay and resurfacing**—unvisited knowledge fades. Revisited knowledge strengthens. SM-2 spaced repetition on all nodes.

7. **Zero-friction capture**—capture action < 2 seconds. Share sheet, voice, screenshot, global shortcut.
8. **Never show a blank page**—always suggest, pre-populate, show context.
9. **Human-in-the-loop by default**—the system proposes, you decide.

17 Implementation Roadmap

17.1 MVP Scope (12–16 Weeks)

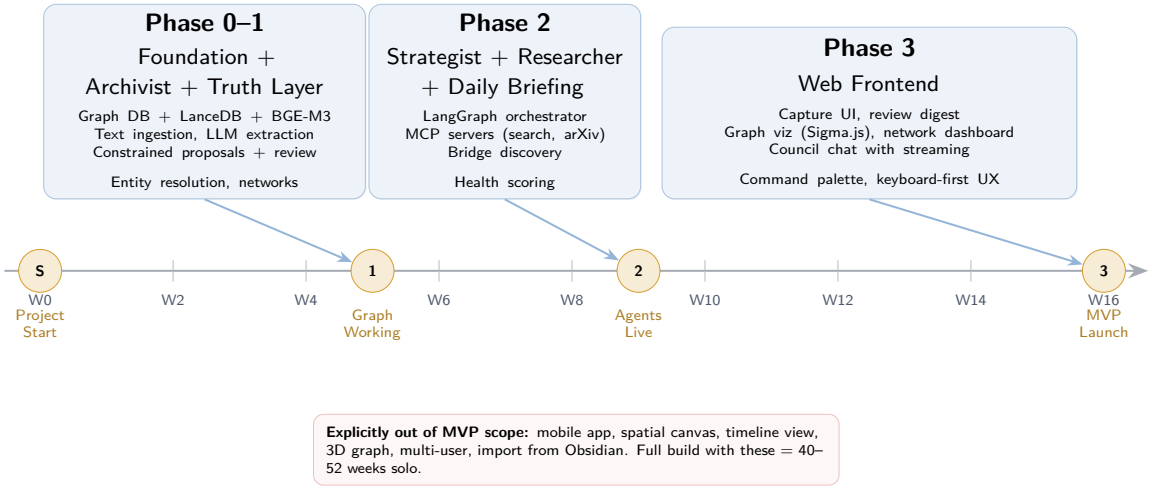

Figure 8: MVP implementation roadmap: 12–16 weeks across three phases with clear milestones. Each phase delivers independently testable value. The scope is deliberately constrained—mobile, spatial canvas, and advanced views are deferred until web proves the concept.

17.2 Cost Structure

Category	Cost	Notes
Infrastructure	\$0/month	Embedded databases, no cloud services
LLM (all Haiku)	~\$4/month	Floor scenario, per user
LLM (Haiku + Sonnet blend)	~\$10–15/month	Recommended routing, per user
LLM (all Sonnet)	~\$15–20/month	Ceiling scenario, per user

Success Metric

Prompt caching reduces Archivist costs by 60–70% (0.1× base cost for cache hits). The Archivist system prompt + graph schema context is near-identical across calls, making it an ideal caching target.

18 Risk Assessment

Risk	Likelihood	Impact	Mitigation
Entity resolution complexity exceeds estimates (3–4 weeks alone)	High	High	Phase 0–1 includes explicit entity resolution scope; prototype early with synthetic data
Auto-approve threshold poorly calibrated, eroding trust	Medium	High	Track correction rate (<5% target); allow user to adjust threshold; daily review digest as safety net
RyuGraph insufficient (Kuzu fork quality unknown)	Medium	Medium	DuckDB fallback

19 Success Metrics

Metric	Target	Why It Matters
Auto-approve correction rate	$< 5\%$	System earns user trust without review fatigue
Daily capture rate	≥ 5 captures/day	Users are actively feeding the graph
Bridge discovery accuracy	$\geq 70\%$ useful	Cross-network connections are the primary value
Daily briefing engagement	$\geq 60\%$ open rate	Users rely on the system for situational awareness
Truth Layer coverage	$\geq 50\%$ of recommendations cite verified facts	Intelligence is grounded, not hallucinated
Graph node revisit rate	$> 33\%$	Breaking the "67% never revisited" baseline
User LLM cost	$<$ \$15/month blended	Sustainable BYOK model

20 Goals & Non-Goals

20.1 Goals

- Build a local-first, privacy-preserving decision intelligence platform that users own completely
- Deliver a self-building knowledge graph from natural language input (text + image)
- Surface cross-domain connections that no single-purpose tool can detect
- Ground all AI outputs in verifiable graph state and sourced facts
- Achieve a sustainable cost model with BYOK LLM APIs ($10–15/month)
- Demonstrate anti-wrapper differentiation: the product survives without any specific LLM provider

20.2 Non-Goals (Explicitly Out of Scope)

- Mobile application (web-first; mobile deferred until concept is proven)
- Calendar sync, bank feeds, location tracking (user is the sole data source)
- Multi-user / collaboration features (single-player first)
- Spatial canvas, timeline view, 3D graph visualization (post-MVP)
- AI companion / chat-as-product (tool, not friend—Dot's failure is the cautionary tale)
- Obsidian vault import (strong adoption hook, but deferred from MVP)

21 Open Questions

1. **Auto-approve threshold tuning**—how aggressive should the default 0.85 threshold be? Need to learn from user correction patterns over time.
2. **Privacy model for MCP research**—when the Researcher agent searches the web, what context does it send? Graph context must be anonymized before external queries.
3. **Obsidian vault import**—importing existing `.md` vaults as a starting point would be a strong adoption hook. Deferred but should be prioritized early post-MVP.
4. **Preventing the collector's fallacy**—decay mechanics + "when did you last reference this?" nudges + synthesis challenges. The system must discourage hoarding and encourage active engagement.

22 Conclusion

Key Insight

Memora is a **decision intelligence platform**, not a note-taking app. It captures life context as input and delivers better high-stakes decisions as output. The knowledge graph, Truth Layer, cross-network bridge discovery, and deterministic algorithms constitute genuine, defensible IP—not an LLM wrapper.

With an MVP timeline of **12–16 weeks**, **zero infrastructure cost**, and a sustainable **$10–15/month** BYOK model, Memora fills six market gaps that no production system currently addresses.

No production system combines local-first privacy, self-building knowledge graphs, cross-domain reasoning, fact-checked intelligence, internet enrichment, temporal knowledge evolution, and proactive decision recommendations. The concept is genuinely novel if executed.

End of Document — Memora Project Proposal v1.0